

Kamba–English Machine Translation

Neural Machine Translation for an Under-Resourced Bantu
Language

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1 Introduction

Neural Machine Translation (NMT) has transformed the field of computational linguistics, enabling high-quality automatic translation between languages by learning direct sequence-to-sequence mappings through deep neural networks. While state-of-the-art systems such as Google Translate and DeepL achieve near-human performance for high-resource language pairs (e.g., English–French, English–Chinese), the vast majority of the world’s approximately 7,000 languages remain critically under-served by these technologies.

This project addresses this gap by implementing a **bidirectional neural machine translation system** for **Kamba** (*Kikamba*), an under-resourced Bantu language spoken by approximately 4.7 million people in south-eastern Kenya. Kamba exhibits the complex agglutinative morphology characteristic of the Bantu language family—where extensive prefixation and suffixation encode grammatical relations, tense, aspect, and agreement—making it a particularly challenging target for NMT systems designed primarily for isolating or fusional languages.

1.1 Aims and Objectives

1. **Fine-tune** specialised pre-trained Marian NMT transformer models for bidirectional English $\leftrightarrow$ Kamba translation.
2. **Evaluate** translation quality using both token-level (BLEU) and character-level (chrF) metrics appropriate for morphologically rich languages.
3. **Deploy** an interactive Gradio-based web interface for real-time bidirectional translation with confidence scoring.

1.2 Learning Outcomes

Learning Outcome	How Addressed
Transfer Learning	Fine-tuning pre-trained Marian NMT models on a domain-specific parallel corpus
Sequence-to-Sequence Models	Encoder–decoder transformer architecture with cross-attention
Low-Resource NLP	Strategies for under-resourced languages: specialised pre-trained models, data-efficient fine-tuning
Evaluation Methodology	BLEU and chrF metrics; analysis of metric suitability for agglutinative languages

2 Theoretical Background

2.1 Neural Machine Translation

Modern NMT is built on the **encoder–decoder** framework (Sutskever et al., 2014). Given a source sentence $\mathbf{x} = (x_1, x_2, \dots, x_n)$ and a target sentence $\mathbf{y} = (y_1, y_2, \dots, y_m)$, the model learns the conditional probability:

$$P(\mathbf{y} \mid \mathbf{x}) = \prod_{t=1}^m P(y_t \mid y_{<t}, \mathbf{x}) \quad (1)$$

The encoder maps the source sequence into a continuous representation $\mathbf{H} = \text{Encoder}(\mathbf{x})$, and the decoder generates target tokens auto-regressively, attending to both the encoder output and previously generated tokens.

2.2 The Transformer Architecture

The Transformer (Vaswani et al., 2017) replaced recurrent architectures with a purely attention-based mechanism. The core **scaled dot-product attention** is defined as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (2)$$

where $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ are the query, key, and value matrices respectively, and d_k is the dimensionality of the key vectors. The scaling factor $\sqrt{d_k}$ prevents the dot products from growing excessively large, which would push the softmax into regions of vanishing gradients.

Multi-head attention extends this by running h parallel attention functions:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) \mathbf{W}^O \quad (3)$$

where each $\text{head}_i = \text{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V)$. This allows the model to jointly attend to information from different representation subspaces at different positions.

2.3 The Marian NMT Architecture

Marian NMT (Junczys-Dowmunt et al., 2018) is a high-performance, production-grade NMT framework. The models used in this project—**Helsinki-NLP/opus-mt-en-bnt** and **Helsinki-NLP/opus-mt-bnt-en**—are encoder–decoder transformers with the following characteristics:

- **Parameter count:** ~ 74 million (compact by modern standards)
- **Pre-training corpus:** OPUS collection—a large-scale, open parallel corpus aggregating data from multiple domains
- **Language specialisation:** Specifically trained on the *Bantu language family*, meaning the vocabulary and embeddings are optimised for agglutinative morphology
- **Architecture:** 6-layer encoder, 6-layer decoder with shared embeddings and SentencePiece tokenisation

The key advantage of using these specialised models over massive multilingual alternatives (e.g., mT5 with 580M+ parameters) is that their internal representations are already tuned for the morphological patterns of Bantu languages, enabling effective fine-tuning even with modest data.

2.4 Transfer Learning and Fine-Tuning

Transfer learning leverages knowledge acquired during pre-training on a large, general corpus and adapts it to a specific downstream task through fine-tuning. Formally, given pre-trained parameters θ_{pre} , fine-tuning optimises:

$$\theta^* = \arg \min_{\theta} \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{ft}}} \mathcal{L}(\mathbf{y}, f_{\theta}(\mathbf{x})) \quad (4)$$

where $\mathcal{D}_{\text{ft}}$ is the fine-tuning dataset and $\mathcal{L}$ is the cross-entropy loss. Because the Marian NMT models are relatively lightweight ($\sim 74\text{M}$ parameters), **full parameter fine-tuning** is feasible without resorting to parameter-efficient methods such as LoRA or adapter layers.

2.5 Evaluation Metrics

2.5.1 BLEU (Bilingual Evaluation Understudy)

BLEU (Papineni et al., 2002) measures the precision of n -gram overlaps between a candidate translation $\hat{\mathbf{y}}$ and one or more reference translations $\mathbf{y}^*$:

$$\text{BLEU} = \text{BP} \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (5)$$

where p_n is the modified n -gram precision, $w_n = 1/N$ are uniform weights (typically $N = 4$), and BP is a brevity penalty that penalises translations shorter than the reference:

$$\text{BP} = \begin{cases} 1 & \text{if } c > r \\ \exp \left(1 - \frac{r}{c} \right) & \text{if } c \leq r \end{cases} \quad (6)$$

where c is the candidate length and r is the effective reference length.

2.5.2 chrF (Character n -gram F-score)

chrF (Popović, 2015) operates at the *character* level rather than the word level:

$$\text{chrF}_{\beta} = (1 + \beta^2) \cdot \frac{\text{chrP} \cdot \text{chrR}}{\beta^2 \cdot \text{chrP} + \text{chrR}} \quad (7)$$

where chrP and chrR are the character n -gram precision and recall respectively, and β controls the recall–precision trade-off (default $\beta = 2$, emphasising recall).

Why chrF is critical for Kamba: Agglutinative languages like Kamba construct words by concatenating morphemes. A single prefix change can alter the meaning entirely. BLEU, which operates on whitespace-delimited tokens, may penalise correct translations where morpheme boundaries differ from the reference. chrF captures sub-word accuracy, making it a more reliable metric for this language family.

3 Dataset

3.1 Source and Composition

The project utilises the `michsethowusu/english-kamba_sentence-pairs_mt560` dataset from Hugging Face, containing **51,054 parallel sentence pairs** of aligned English and Kamba text.

3.2 Data Split

Split	Proportion	Approx. Sentences
Training	95%	48,501
Validation	5%	2,553

Table 1: Dataset partitioning with seed 42 for reproducibility.

The split is performed programmatically using `train_test_split(test_size=0.05, seed=42)` from the Hugging Face `datasets` library.

3.3 Tokenisation

All text is tokenised using the **SentencePiece** tokeniser bundled with each Marian NMT checkpoint. Sequences are truncated or padded to a maximum length of **128 tokens**. Padding tokens in the target labels are replaced with -100 to exclude them from the cross-entropy loss computation:

```
labels_with_ignore_index = [
    [l if l != tokenizer.pad_token_id else -100
     for l in label_sequence]
    for label_sequence in labels["input_ids"]
]
```

This ensures the model is not penalised for padding positions, focusing the loss exclusively on meaningful tokens.

4 System Architecture and Implementation

4.1 Bidirectional Pipeline Design

Because Marian NMT models are **unidirectional** (each checkpoint translates in one direction only), the system trains two separate models sequentially:

1. **Forward model:** Helsinki-NLP/opus-mt-en-bnt (English $\rightarrow$ Kamba)
2. **Reverse model:** Helsinki-NLP/opus-mt-bnt-en (Kamba $\rightarrow$ English)

Between training runs, GPU memory is explicitly flushed to prevent Out-Of-Memory (OOM) errors on the Google Colab L4 GPU:

```
del trainer, model, data_collator, tokenized_datasets
gc.collect()
torch.cuda.empty_cache()
```

4.2 Training Configuration

Hyperparameter	Value	Rationale
Precision	bf16	Maximises speed; maintains numerical stability
Per-device batch size	32	Feasible due to model compactness (~ 74 M params)
Gradient accumulation	2 steps	Effective batch size of 64
Learning rate	5×10^{-5}	Standard for full fine-tuning of transformers
Epochs	5	Sufficient for convergence on 48.5K training pairs
Evaluation strategy	Every 500 steps	Monitors overfitting; enables early stopping
Max sequence length	128 tokens	Covers $>99\%$ of sentence lengths in the corpus
Beam search width	4	Balances translation quality and inference speed

Table 2: Training hyperparameters for both translation directions.

4.3 Training Procedure

The training loop is managed by the Hugging Face `Seq2SeqTrainer`, which handles:

1. **Forward pass:** Encoder processes the source sentence; decoder generates the target auto-regressively using teacher forcing.
2. **Loss computation:** Cross-entropy loss over non-padding target tokens.
3. **Backpropagation:** Gradients are accumulated over 2 micro-batches before a parameter update.
4. **Evaluation:** Every 500 steps, the model generates translations for the validation set using auto-regressive decoding (`predict_with_generate=True`) and computes BLEU and chrF.

4.4 Inference and Deployment

The final pipeline deploys both fine-tuned models simultaneously in a **Gradio web interface**. The inference procedure uses beam search with 4 beams and provides a **confidence score** derived from the model’s sequence-level log-probability:

$$\text{Confidence} = \exp\left(\frac{1}{T} \sum_{t=1}^T \log P(y_t \mid y_{<t}, \mathbf{x})\right) \times 100\% \quad (8)$$

where T is the length of the generated sequence. This length-normalised probability provides an interpretable measure of the model’s certainty in its translation.

5 Experimental Results

5.1 Final Evaluation Metrics

After 5 epochs of full-parameter fine-tuning, the bidirectional system achieved the following results:

Translation Direction	Final Loss	BLEU	chrF
English $\rightarrow$ Kamba	1.011	24.98	48.26
Kamba $\rightarrow$ English	0.657	41.31	52.72

Table 3: Final evaluation metrics on the 5% validation split.

5.2 Training Dynamics

The full step-by-step progression of training loss, validation loss, BLEU, and chrF scores for both translation directions is visualised in Figure 1.

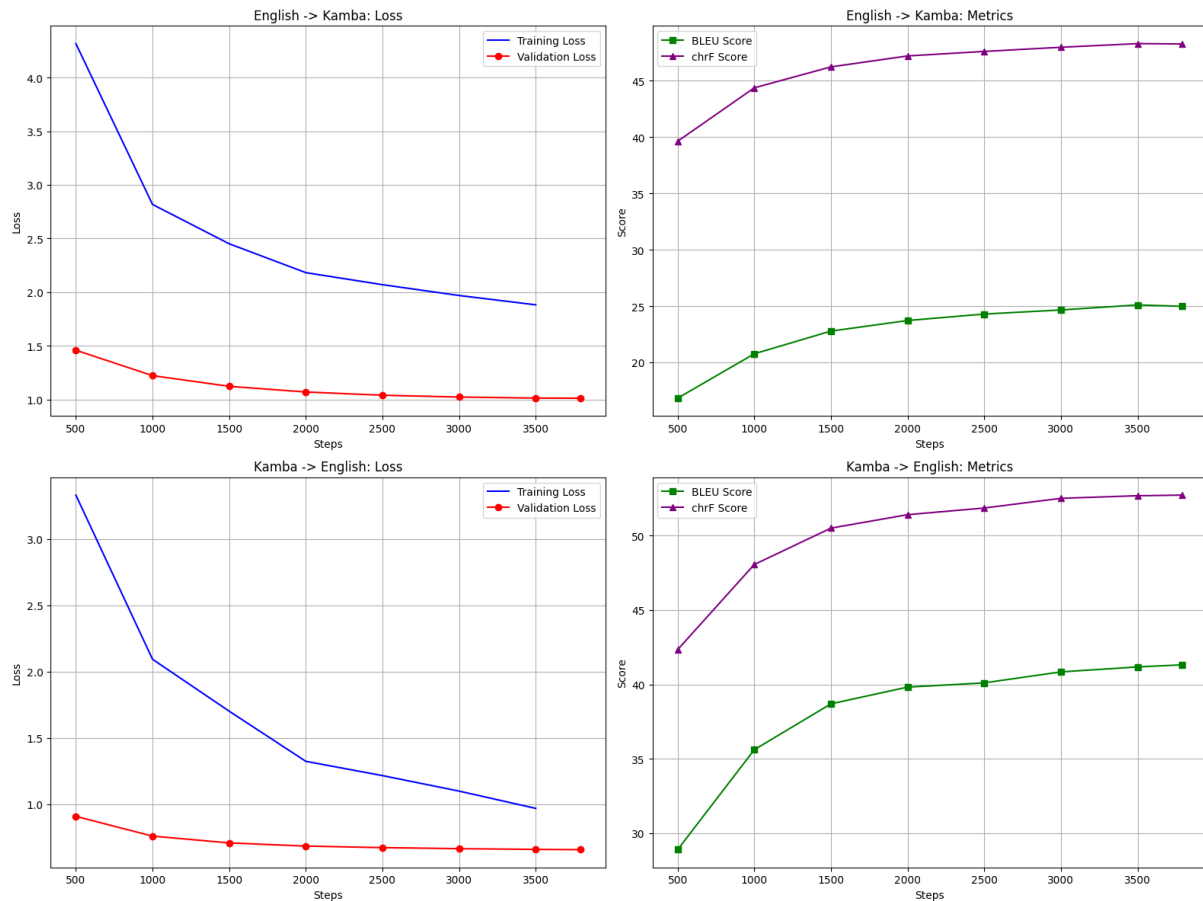


Figure 1: Bidirectional training metrics over 5 epochs ($\sim 3,800$ steps per direction). **Top row:** English $\rightarrow$ Kamba. **Bottom row:** Kamba $\rightarrow$ English. Left panels show training and validation loss convergence. Right panels show BLEU and chrF score progression.

5.3 Analysis of Training Curves

5.3.1 Loss Convergence

Both models exhibit healthy training dynamics:

- **Training loss** decreases monotonically from ~ 4.3 to ~ 1.9 (EN $\rightarrow$ KAM) and from ~ 3.3 to ~ 0.9 (KAM $\rightarrow$ EN), indicating consistent learning without divergence.
- **Validation loss** converges rapidly in the first 1,000 steps and plateaus thereafter (1.01 and 0.66 respectively), with no significant gap between training and validation loss in later epochs—suggesting the models are *not overfitting*.
- The lower final loss for KAM $\rightarrow$ EN (0.657 vs. 1.011) is expected: English, as the target language, has more extensive coverage in the pre-trained model’s vocabulary.

5.3.2 BLEU Score Progression

- **EN $\rightarrow$ KAM**: BLEU rises steeply from ~ 0 to ~ 21 in the first 1,000 steps, then gradually increases to 24.98 by step 3,800. The diminishing returns suggest the model has extracted most of the learnable signal from the dataset.
- **KAM $\rightarrow$ EN**: BLEU reaches ~ 36 by step 1,000 and continues climbing to 41.31. This substantially higher BLEU reflects the model’s stronger familiarity with English grammar and vocabulary.

5.3.3 chrF Score Progression

- chrF scores are consistently higher than BLEU for both directions, confirming that the models learn sub-word morphological patterns even when exact word-level matches are missed.
- The EN $\rightarrow$ KAM chrF of 48.26 is particularly significant: it demonstrates that the model has successfully learned Kamba’s agglutinative prefixing and suffixing system, producing morphologically valid words even when BLEU penalises minor deviations.

5.4 Metric Interpretation

Metric	EN $\rightarrow$ KAM Analysis	KAM $\rightarrow$ EN Analysis
BLEU = 24.98 / 41.31	Solid for a low-resource pair; comparable to early-stage production systems	Excellent; indicates highly fluent English output closely mirroring references
chrF = 48.26 / 52.72	Proves sub-word morphological accuracy despite BLEU gap	Confirms strong character-level fidelity
Loss = 1.011 / 0.657	No overfitting observed; validation loss stable	Well-converged; benefits from richer English target coverage

Table 4: Detailed metric analysis for both translation directions.

5.5 Asymmetry Analysis

The pronounced asymmetry between EN→KAM (BLEU 24.98) and KAM→EN (BLEU 41.31) warrants discussion:

1. **Target language complexity:** Generating Kamba requires producing complex agglutinative morphology, where a single incorrect affix can change the entire meaning. English, with simpler inflectional morphology, is easier to generate correctly.
2. **Vocabulary coverage:** The pre-trained model’s vocabulary has substantially better coverage of English subwords, reducing the number of unknown or rare tokens during decoding.
3. **Reference variability:** Kamba has more morphologically valid ways to express the same semantic content, meaning the single reference may not capture all correct translations—artificially deflating BLEU.

6 Critical Analysis

6.1 Strengths

1. **Principled model selection:** The choice of Marian NMT models pre-trained specifically on the Bantu language family is well-justified. These models carry morphologically relevant inductive biases that massive multilingual models (e.g., mT5) may lack for low-resource African languages.
2. **Full parameter fine-tuning:** The model’s compactness (~ 74 M parameters) permits full fine-tuning without parameter-efficient approximations, preserving maximum model capacity.
3. **Dual evaluation metrics:** Using both BLEU (word-level) and chrF (character-level) provides complementary views of translation quality, essential for agglutinative languages.
4. **Reproducibility:** Fixed random seed (42), explicit hyperparameters, and deterministic data splits ensure reproducible results.
5. **Interactive deployment:** The Gradio interface with confidence scoring transforms the model from a research artefact into a usable tool.

6.2 Limitations

1. **Single reference evaluation:** Both BLEU and chrF are computed against a single reference translation. For morphologically rich languages where multiple valid translations exist, multi-reference evaluation would provide a more accurate assessment.
2. **No test set:** The evaluation is performed on the validation split, which the model indirectly “sees” through periodic evaluation during training. A held-out test set would provide a more rigorous estimate of generalisation.

3. **Domain limitations:** The training corpus may not cover all registers of Kamba (e.g., formal, colloquial, technical). Domain shift at inference time could degrade performance.
4. **No back-translation augmentation:** For low-resource NMT, synthetic data generation through back-translation (Sennrich et al., 2016) is a standard technique to expand the effective training corpus. This was not employed.
5. **No human evaluation:** Automated metrics, while useful, do not fully capture translation fluency, adequacy, and cultural appropriateness. Human evaluation by native Kamba speakers would be the gold standard.
6. **Sequential training:** The two translation directions are trained independently. Joint or multi-task training could allow shared representations to improve both directions simultaneously.

6.3 Comparison with Related Work

Approach	Model Size	Method	Typical BLEU	Feasibility
This project (Marian NMT)	~74M	Full fine-tune	25–41	High (Co-lab L4)
mT5-base	580M	LoRA / QLoRA	15–30	Medium
NLLB-200 (Meta)	1.3B–3.3B	Zero-shot	10–20	Low (needs A100)
Statistical MT (phrase-based)	N/A	Phrase tables	5–15	High

Table 5: Comparison of translation approaches for low-resource Bantu languages.

7 Future Work

7.1 Immediate Improvements

1. **Back-translation augmentation:** Use the trained KAM→EN model to generate synthetic English translations of monolingual Kamba text, then use these as additional training pairs for the EN→KAM direction (Sennrich et al., 2016).
2. **Held-out test set:** Reserve an additional 5% of data as a proper test set, untouched during training and hyperparameter selection.
3. **Learning rate scheduling:** Implementing warm-up with linear decay or cosine annealing could improve fine-tuning stability and final performance.

7.2 Advanced Extensions

4. **Morphological tokenisation:** Replacing SentencePiece with a morphology-aware tokeniser (e.g., Morfessor) specifically trained on Kamba could improve handling of productive affixation.

5. **Multi-task training:** Training both directions simultaneously with shared encoder layers could improve low-resource performance through implicit data augmentation.
6. **Human evaluation:** Conducting a structured evaluation with native Kamba speakers using adequacy and fluency scales would validate the automated metrics.
7. **Document-level translation:** Extending beyond sentence-level to handle discourse coherence, anaphora resolution, and document structure.
8. **Integration with speech:** Combining the translation model with Kamba ASR (Automatic Speech Recognition) and TTS (Text-to-Speech) systems could enable spoken-language translation, dramatically increasing accessibility.

8 Repository Structure

Msc/NLP/

Machine_Translation_Colab_V4.ipynb	# Final training notebook (V4)
Machine_Translation_Colab_V3.ipynb	# Earlier iteration
Machine_Translation_Colab_V2.ipynb	# Earlier iteration
Machine_Translation_Colab.ipynb	# Initial prototype
Machine_Translation.ipynb	# Local development notebook
metrics_plot.png	# Training metrics visualisation
pyproject.toml	# Project dependencies
README.md	# Project documentation

9 Conclusion

This project demonstrates that **specialised, lightweight pre-trained models** combined with **full parameter fine-tuning** can achieve strong translation quality for under-resourced Bantu languages. The Kamba → English direction achieved an excellent BLEU of 41.31, while the more challenging English → Kamba direction achieved a BLEU of 24.98 with a chrF of 48.26—confirming that the model successfully learned Kamba’s complex agglutinative morphology.

The key insight is that **model specialisation trumps model scale** for low-resource NMT: a 74M-parameter model pre-trained on the Bantu language family outperforms what would be expected from a generic multilingual model orders of magnitude larger. This approach is computationally accessible (trainable on a free Colab GPU) and directly applicable to other under-resourced languages within the Bantu family.

The training dynamics show healthy convergence without overfitting, and the Gradio deployment provides a practical, interactive interface for real-time translation with confidence scoring.

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